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JAVA PROJECT REPORT ON

SPORTS TOURNAMENT TEAM REGISTRATION AND MATCH RESULT SYSTEM

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DECLARATION

I hereby declare that the project entitled "**SPORTS TOURNAMENT TEAM REGISTRATION AND MATCH RESULT SYSTEM**" submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of **MUGILAN R**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(PREM S B)

CERTIFICATE

This is to certify that the project entitled **"SPORTS TOURNAMENT TEAM REGISTRATION AND MATCH RESULT SYSTEM"** is a **bona fide work** carried out by **PREM S B , B.Sc.IT-'B'**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **JAVA PROGRAMMING**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date: 09.04.2026

Submitted for the Viva-Voce Examination held on _____

Guide:
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ABSTRACT

The **Sports Tournament Team Registration and Match Result System** is a Java-based application developed to simplify the management of sports tournaments. In many schools, colleges, and local competitions, tournament details such as team registrations, match schedules, and match results are recorded manually on paper or spreadsheets. These traditional methods often lead to errors, missing information, difficulty in updating results, and overall inefficiency in organizing events.

This project provides an automated solution that allows organizers to register teams, create match schedules, record match results, and view all tournament information in an organized format. The system is built using Core Java concepts such as classes, objects, ArrayLists, loops, conditional statements, and user-defined methods. The menu-driven interface makes it easy for users to perform operations like adding teams, generating matches, and entering results.

The main objective of this system is to help tournament coordinators manage event data efficiently while reducing manual work and errors. The application ensures smooth handling of team details and match outcomes, making the tournament management process faster, more accurate, and more organized.

This project demonstrates the practical use of Java programming in developing real-world sports management applications and serves as a foundation for building more advanced tournament systems with features such as ranking tables, multiple rounds, and database integration.

1. INTRODUCTION

The **Sports Tournament Team Registration and Match Result System** is a Java-based application designed to simplify the process of managing sports tournaments. The system helps organizers register teams, maintain match details, record match results, and display rankings in a structured and efficient manner.

Traditionally, tournament data is maintained on paper or spreadsheets, which often leads to confusion, missing records, and difficulty in updating match results. This system provides a

digital solution where team details, match schedules, and match results can be easily stored and managed.

The system uses Core Java concepts such as classes, objects, ArrayLists, loops, conditional statements, and methods. It can be developed as a simple console-based program suitable for academic projects.

The main purpose of the system is to help tournament coordinators handle registrations and results in an organized manner while demonstrating Java programming skills through a real-life application.

1.1 Objective

The primary objective of this system is to automate the registration and result management of sports tournaments.

Specific objectives:

•To provide a platform for team registration.

•To maintain team and match details digitally.

•To record match results easily.

•To allow organizers to view teams and results in an organized list.

•To reduce manual errors and paper-based work.

•To demonstrate Java concepts such as OOP, lists, loops, and conditional logic.

1.2 Overview

The **Sports Tournament Team Registration and Match Result System** allows organizers to:

•Register teams with details such as team name and captain name.

•Create match records.

Update match results (winning team, scores).

View all registered teams and match outcomes.

•The program uses a dynamic ArrayList to store team and match information and provides a clear, menu-driven interface.

1.3 Features

•Team Registration – Add new teams with basic details.

•Match Creation – Enter match schedule and participating teams.

•Result Entry – Record match winner and score.

•View Teams – Display list of all registered teams.

•View Match Results – Display results of completed matches.

•Simple Menu System – Easy-to-use menu-driven interface.

•Error Handling – Prevents invalid team or match selections.

2. PROGRAM ANALYSIS

Program analysis helps understand the challenges in managing tournaments manually and identifies the requirements of an automated sports management system.

Tournament organizers often face issues such as misplaced team registration forms, incorrect match results, difficulty preparing rankings, and confusion in schedules. By analyzing these problems, the need for a simple automated system becomes clear.

2.1 Problem Definition

Organizers using manual methods face problems such as:

•Difficulty tracking teams and match schedules.

•Mismanagement or loss of registration details.

•Errors while updating match results.

•No proper method to view results quickly.

•Confusion in ranking teams.

Time-consuming paperwork.

A digital system is required to simplify registration and record match results efficiently.

2.2 Project Overview

This project provides an automated solution for sports tournament management.

The main functions include:

•Team registration

•Recording match details

•Updating match results

•Viewing teams and match outcomes

It is developed using Core Java concepts such as classes, objects, ArrayList, and methods.

2.3 Scopes and Limitations

Scope:

The system can be used for:

School tournaments

•College fests

•Small sports leagues

•Local competitions

•Indoor and outdoor sports

Limitations:

•No online access or cloud storage

•No automatic ranking calculation

•No notifications or scheduling automation

•Console-based (no GUI)

3. SYSTEM DESIGN

System design defines the structure of the application and how each module interacts.

The program is divided into smaller parts such as registration, match creation, and result entry.

ArrayLists are used to store team and match data, making it easier to add or update information.

3.1 Module

1.Team Registration Module

•Accepts team name and captain name

•Stores team details in the list

2.Team Display Module

•Shows all registered teams with IDs

3.Match Creation Module

•Allows selection of two teams

•Creates a match entry

4.Match Result Module

•Records the winning team and match score

•Updates match outcome

5.Match Display Module

•Shows completed match results

6.Menu and Navigation Module

•Provides options such as register team, add match, enter result, view teams, exit

7.Exit Module

Safely ends the program

4. METHODOLOGY

The project follows steps similar to your template:

4.1 Problem Analysis

Manual tournament management problems are studied and requirements are identified—team registration, storing match data, result recording.

4.2 System Design

Modules are created for team registration, match handling, and result updates.

ArrayLists and classes are chosen for storing data.

4.3 Program Development

Coding is done in Java using loops, conditional statements, objects, and lists.

A clear menu-driven interface is created.

4.4 Testing and Validation

Test cases check:

•Adding teams

•Creating matches

•Entering results

•Handling invalid team IDs

•Viewing results

4.5 Result and Documentation

Screens and outputs are documented.

The system shows successful registration and match result updates.

4.6 Algorithm

Basic steps:

1.Start

2.Display menu

3.Get user choice

4.Perform team or match operation

5.Show output

6.Repeat until exit

4.7 Flowchart

Start

↓

Menu

↓

User choice

↓

Operation

↓

Display output

↓

Repeat

↓

End

5. SOURCE CODE

```
import java.util.*;

class Team {
    String name, captain;
```

```
    Team(String name, String captain) {
        this.name = name;
        this.captain = captain;
    }
}

class Match {
    String team1, team2, winner;
    String score;

    Match(String t1, String t2) {
        team1 = t1;
        team2 = t2;
    }
}

public class TournamentSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<Team> teams = new ArrayList<>();
        ArrayList<Match> matches = new ArrayList<>();

        int choice;

        do {
            System.out.println("\n1. Register Team\n2. View Teams\n3. Create Match\n4. Enter Result\n5. View Results\n6. Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt();
```

```
            sc.nextLine();

            if (choice == 1) {
                System.out.print("Team Name: ");
                String name = sc.nextLine();
                System.out.print("Captain Name: ");
                String captain = sc.nextLine();
                teams.add(new Team(name, captain));
                System.out.println("Team Registered!");

            } else if (choice == 2) {
                System.out.println("\n--- Registered Teams ---");
                for (int i = 0; i < teams.size(); i++) {
                    System.out.println((i + 1) + ". " + teams.get(i).name + " (Captain: " +
teams.get(i).captain + ")");
                }

            } else if (choice == 3) {
                System.out.print("Enter Team 1 ID: ");
                int t1 = sc.nextInt();
                System.out.print("Enter Team 2 ID: ");
                int t2 = sc.nextInt();

                if (t1 <= teams.size() && t2 <= teams.size() && t1 != t2) {
                    matches.add(new Match(teams.get(t1 - 1).name, teams.get(t2 - 1).name));
                    System.out.println("Match Created!");
                } else {
                    System.out.println("Invalid Team IDs!");
                }
```

```
    } else if (choice == 4) {
        System.out.print("Enter Match Number: ");
        int m = sc.nextInt();
        sc.nextLine();

        if (m <= matches.size()) {
            Match match = matches.get(m - 1);
            System.out.print("Winner: ");
            match.winner = sc.nextLine();
            System.out.print("Score: ");
            match.score = sc.nextLine();
            System.out.println("Result Updated!");
        } else {
            System.out.println("Invalid Match Number!");
        }

    } else if (choice == 5) {
        System.out.println("\n--- Match Results ---");
        for (int i = 0; i < matches.size(); i++) {
            Match mt = matches.get(i);
            System.out.println((i + 1) + ". " + mt.team1 + " vs " + mt.team2 +
                    " | Winner: " + mt.winner + " | Score: " + mt.score);
        }
    }

} while (choice != 6);

System.out.println("Exited.");
```

```
    }
}
```

6. INPUT AND OUTPUT SCREENS

6. INPUT & OUTPUT SCREENS

Sample screens:

•Team registration screen

•Team list display

•Match creation screen

•Result entry screen

•Match result display

7. DISCUSSION & CONCLUSION

7.1 Discussion

The system simplifies tournament management by allowing easy team registration and match result recording. Java features such as classes, ArrayLists, methods, loops, and conditionals were used to build a modular and user-friendly application.

Testing confirmed that the system correctly registers teams, creates matches, and updates results without errors.

7.2 Conclusion

The project successfully demonstrates a working sports tournament management system. It achieves the goal of digitalizing registrations and results and reduces manual errors. The system can be expanded with features like leaderboards, multiple rounds, GUI, and database storage in future versions

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com